

Computer Networking Notes

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About This Notes

This document is part of my personal Computer Networking learning journey. These notes are organized to help me build a strong foundation in networking through theory, practical labs, and real-world examples. They are intended for educational purposes and may be updated as my knowledge grows.

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CIDR (Classless Inter-Domain Routing) Notation

CID (Classless Inter Domain Routing) is a way to represent an IP address and its subnet mask in a shorter and easier format.

Instead of writing:

- IP Address: 192.168.1.0
- Subnet Mask: 255.255.255.0

We write:

- 192.168.1.0/24

The /24 is called the CIDR notation or prefix length.

What does /24 mean?

The number after the slash tells you:

- How many bits are used for the Network ID.

Remember

- An IPv4 address has 32 bits.

Example:

- 192.168.1.0/24

means

- Network bits = 24
- Host bits = 8

Because

- $32 - 24 = 8$

CIDR and Subnet Mask

CIDR	Subnet Mask	Total IPs	Usable IPs
/32	255.255.255.255	1	1
/31	255.255.255.254	2	0
/30	255.255.255.252	4	2
/29	255.255.255.248	8	6
/28	255.255.255.240	16	14
/27	255.255.255.224	32	30
/26	255.255.255.192	64	62
/25	255.255.255.128	128	126
/24	255.255.255.0	256	254
/23	255.255.254.0	512	510
/22	255.255.252.0	1024	1022
/21	255.255.248.0	2048	2046
/20	255.255.240.0	4096	4094
/19	255.255.224.0	8192	8190
/18	255.255.192.0	16,384	16,382
/17	255.255.128.0	32,768	32,766
/16	255.255.0.0	65,536	65,534
/15	255.254.0.0	131,072	131,070
/14	255.252.0.0	262,144	262,142
/13	255.248.0.0	524,288	524,286
/12	255.240.0.0	1,048,576	1,048,574
/11	255.224.0.0	2,097,152	2,097,150
/10	255.192.0.0	4,194,304	4,194,302
/9	255.128.0.0	8,388,608	8,388,606
/8	255.0.0.0	16,777,216	16,777,214
/7	254.0.0.0	33,554,432	33,554,430
/6	252.0.0.0	67,108,864	67,108,862
/5	248.0.0.0	134,217,728	134,217,726
/4	240.0.0.0	268,435,456	268,435,454
/3	224.0.0.0	536,870,912	536,870,910
/2	192.0.0.0	1,073,741,824	1,073,741,822
/1	128.0.0.0	2,147,483,648	2,147,483,646
/0	0.0.0.0	4,294,967,296	4,294,967,294

Image Credit: <https://www.ad-net.com.tw/subnet-chart-cidr-values/>

Why CIDR is Used

Before CIDR, networks used fixed classes (Class A, B, and C). This often wasted IP addresses because organizations had to take an entire class even if they needed far fewer addresses.

CIDR solves this by allowing networks to be sized more precisely. For example

- /24 → 254 usable hosts
- /26 → 62 usable hosts
- /28 → 14 usable hosts

This flexibility conserves IPv4 addresses and enables more efficient routing.

Practice Questions

- What is the subnet mask for 192.168.1.0/27?
- How many usable hosts are available in a /29 network?
- How many host bits does a /22 network have?
- What is the CIDR notation for the subnet mask 255.255.255.240?
- What is the subnet mask for a /30 network?